

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I P.Karpaga shree(192424042) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Karpaga shree

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Name : P. Karipaga shree

Test : 2

Register number : 192424042

date : 22/08/2026

course code : CSA1403

1) Given statement :

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

given: if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

conditions: $a < b$

$c > d$

$a > d$

Labels: L1, L2, L3, L4, L5

L1:

if $(a < b)$ goto L2

goto L5

L2: if $(c < d)$ goto L4

goto L3

L3: if $(a > d)$ goto L4

goto L5

Three address code :

38/40

L4 $\Rightarrow z = x + y * z$

$\therefore L4 :$

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L6

L5 $\Rightarrow z = z + 1$

$\therefore L5 :$

$t_3 = z + 1$

$z = t_3$

② Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using Back patching.

$[(a < b) \text{ and } (c < d)] \text{ or } (e < f)$

1. $a < b$ goto 3

2. goto 5

3. $c < d$ goto 7

4. goto 5

5. $e < f$ goto 7

6. goto 8

7. true

8. false

③ construct consider the arithmetic expression:

$$A = (-A + B) * ([-A + B])$$

three address code:

$$t_1 = -A$$

$$t_2 = t_1 + B$$

$$t_3 = -A$$

$$t_4 = t_3 + B$$

$$t_5 = t_4 * c$$

$$t_6 = t_2 * t_5$$

$$A = t_6$$

Quadruples:

	op	arg(1)	arg(2)	result
(0)	unifus	A	-	t ₁
(1)	+	t ₁	B	t ₂
(2)	unifus	A	-	t ₃
(3)	+	t ₃	B	t ₄
(4)	*	t ₄	c	t ₅
(5)	*	t ₂	t ₅	t ₆
(6)	=	t ₆	-	t ₇

④ construct three address code using the following production for boolean expression $(a > b)$ or $(c > d)$ or $(e < f)$

$$B \rightarrow B_1 \parallel B_2$$

$$B \rightarrow B_1 \&\& B_2$$

$$B \rightarrow !B_1$$

$$B \rightarrow E_1 \text{ relop } E_2$$

$$B \rightarrow \text{true}$$

$$B \rightarrow \text{false}$$

4:

if $a > b$ goto LTRUE

goto LE

LE: if $(c < d)$ goto LTRUE

goto LFALSE L3

L3: if $(e < f)$ goto LTRUE

goto LFALSE

LTRUE:

// Boolean expression is true

goto Lend

Lend:

LFALSE:

// Boolean expression
is false